Chinmay Mysore Dharmendra

Curriculum Vitae
Department of Mathematics, University of Connecticut, Storrs, CT
chinmaymd@uconn.edu

EDUCATION

Ph.D. in Mathematics, University of Connecticut 2021 – (Expected May 2027)

- Advisor: Tom Roby
- Dissertation: Combinatorics of Okada arc diagrams
- Area: Algebraic Combinatorics

Master of Mathematics (M.Math.), Indian Statistical Institute, Bangalore 2019 – 2021

- First Division with Distinction
- Master's project: *Structure of Linear Groups and Growth Functions* (advisor: Prof. B. Sury)

Bachelor of Mathematics (Honours) (B.Math.), Indian Statistical Institute, Bangalore 2016 – 2019

- First Division with Distinction

RESEARCH INTERESTS

Dynamical algebraic combinatorics (homomesy, combinatorial dynamics); representation theory and differential posets; pattern avoidance and Catalan objects; computational and experimental combinatorics.

CURRENT RESEARCH PROJECTS

Homomesy of Linear Feedback Shifts

- Studying homomesy (constant orbit averages) for masked and affine Fibonacci and Galois linear feedback shift maps. Manuscript in preparation.

Okada Arc Diagrams and the Young–Fibonacci Lattice

- Investigating Okada arc diagrams and the Hivert–Scott RSK-type bijection associated with the Young–Fibonacci lattice, a differential poset.

PUBLICATIONS AND PREPRINTS

- C. M. Dharmendra, M. Plante, T. Roby, *Homomesies of Linear Feedback Shifts*, in preparation.

TALKS AND PRESENTATIONS

Invited and Contributed Talks

- Homomesies of Linear Feedback Shifts — SIGMA Seminar, UConn Oct 2026
- Differential Posets and the Algebras Associated to Them — SIGMA Seminar, UConn Apr 2025
- Homomesies of Linear Feedback Shifts — Graduate Online Combinatorics Colloquium Apr 2025
- Homomesies of Linear Feedback Shifts — AMS Spring Eastern Sectional Meeting Apr 2025
- Homomesies of Linear Feedback Shifts — SIGMA Seminar, UConn Oct 2024

Posters

- Homomesies of Linear Feedback Shifts — MAAGC 2024 (Mid-Atlantic Algebra, Geometry & Combinatorics) Oct 2024

FELLOWSHIPS, AWARDS, AND HONORS

- Teaching Innovation Fellow (Multivariable Calculus), University of Connecticut — AY 2026–27
- EMITT Fellow (Equity-Minded Inquiry & Transformative Teaching), UConn — AY 2025–26
- Summer Travel Fellowship, UConn — 2024, 2026
- INSPIRE-SHE Scholarship, Department of Science and Technology, Government of India — 2016 – 2021
- Simon Marais Mathematics Competition — Top Quartile (Pairs Division) — 2018

TEACHING EXPERIENCE

All positions at the University of Connecticut, Department of Mathematics.

Instructor of Record

- MATH 2110Q Multivariable Calculus (Teaching Innovation Fellow) — Fall 2026
- MATH 1060Q Precalculus — Fall 2025, Spring 2026
- MATH 2410Q Elementary Differential Equations — Spring 2024

Full responsibility for lectures, course materials, assessments, and grading.

Graduate Supervisor, Q Tutoring Center — Spring 2024, Fall 2024

- Supervised tutoring for mathematics Q courses
- Interviewed and hired undergraduate mathematics tutors
- Led pedagogy workshops to train tutors

Teaching Assistant

- MATH 1070Q Mathematics for Business & Economics — Fall 2021, Spring 2022, Summer 2023, Fall 2023
- MATH 1030Q Elementary Discrete Mathematics — Spring 2023
- MATH 2210Q Multivariable Calculus — Fall 2022

CURRICULUM AND PEDAGOGICAL DEVELOPMENT

- Designed staged, team-based active-learning activities for Multivariable Calculus, including a contour-plot analysis of gradients and directional derivatives and a triple-integral activity on reordering the order of integration.
- Built question banks for assessments in financial mathematics (simple and compound interest, effective annual rate, amortization, sinking funds, annuities), linear programming, linear algebra, combinatorics, and probability for MATH 1070Q.
- Participated in the EMITT program on equity-minded inquiry and transformative teaching (AY 2025–26).

MENTORING

- Directed Reading Program, University of Connecticut — mentored undergraduate Carter Gay in a reading of Wilf's *generatingfunctionology* — Fall 2025

COMPUTATIONAL SKILLS

SageMath, Python (including matplotlib), C

LANGUAGES

English (fluent), Kannada (fluent)